

Class work 7

Task 1. Construct vector notation of the function h by given vector notations of functions f and g

- (a) $f = (0001), g = (1101), h(x, y, z) = f(y, g(x, z)) \vee g(y, z)$
- (b) $f = (0111), g = (1101), h(x, y, z) = f(g(x, y), x) \& g(x, z)$
- (c) $f = (1001), g = (1101), h(x, y, z) = f(x, y) + g(f(x, x), z)$
- (d) $f = (0110), g = (1101), h(x, y, z) = f(x, z) \sim g(y, f(x, x))$
- (e) $f = (1110), g = (1101), h(x, y, z) = f(x, f(x, x)) \rightarrow g(y, z)$
- (f) $f = (1000), g = (1101), h(x, y, z) = f(f(y, y), x) \downarrow g(y, z)$
- (g) $f = (1011), g = (1101), h(x, y, z) = f(x, z) \mid g(g(y, z), z)$
- (h) $f = (0010), g = (1101), h(x, y, z) = f(z, x) \vee g(y, g(z, z))$
- (i) $f = (0100), g = (1101), h(x, y, z) = f(z, g(y, x)) \& g(y, y)$
- (j) $f = (0001), g = (0100), h(x, y, z) = f(g(y, z), x) + g(x, y)$
- (k) $f = (0111), g = (0100), h(x, y, z) = f(z, x) \rightarrow g(f(y, z), x)$
- (l) $f = (1001), g = (0100), h(x, y, z) = f(y, f(z, y)) \downarrow g(x, z)$
- (m) $f = (0110), g = (0100), h(x, y, z) = f(f(x, z), z) \mid g(z, y)$
- (n) $f = (1000), g = (0100), h(x, y, z) = f(y, x) \vee g(g(x, x), z)$
- (o) $f = (0010), g = (0100), h(x, y, z) = f(z, z) \& g(z, g(x, y))$
- (p) $f = (1110), g = (0100), h(x, y, z) = f(x, g(z, y)) + g(z, x)$
- (q) $f = (0111), g = (1110), h(x, y, z) = f(g(z, x), y) \sim g(y, x)$
- (r) $f = (1001), g = (1110), h(x, y, z) = f(z, f(y, z)) \rightarrow g(z, x)$
- (s) $f = (0110), g = (1110), h(x, y, z) = f(y, z) \downarrow g(f(z, y), x)$
- (t) $f = (1000), g = (1110), h(x, y, z) = f(z, x) \mid g(g(z, z), y)$
- (u) $f = (0001), g = (1110), h(x, y, z) = f(x, g(x, x)) \vee g(y, z)$

(v) $f = (0010), g = (1110), h(x, y, z) = f(f(y, x), z) \& g(x, x)$

(w) $f = (0001), g = (0110), h(x, y, z) = f(x, y) + g(z, g(z, z))$

(x) $f = (0111), g = (0110), h(x, y, z) = f(y, g(y, z)) \sim g(x, y)$

(y) $f = (1110), g = (0110), h(x, y, z) = f(g(x, y), z) \rightarrow g(z, y)$

(z) $f = (1000), g = (0110), h(x, y, z) = f(x, f(z, x)) \downarrow g(x, y)$

Task 2. Construct the Truth Table of the following function

(a) $(x \mid y) \vee yz$

(b) $((x + y) \sim y) \mid z$

(c) $x \vee (yz + x)$

(d) $x \downarrow (y \downarrow (y \mid z))$

(e) $((x + y) \mid x)z$

(f) $(x \sim y) \vee (z \downarrow y)$

(g) $((x \vee y) \sim y) + z$

(h) $x \vee (y \downarrow xz)$

(i) $(x \vee y) \sim (z \downarrow x)$

(j) $(x + y) \downarrow (z \vee x)$

(k) $(x \mid yz)x$

(l) $(x \rightarrow y) \downarrow (x + z)$

(m) $x(x \downarrow y)z$

(n) $x \sim ((y \mid z) + x)$

(o) $x \mid (yz + z)$

(p) $x \downarrow ((y \sim z) \downarrow x)$

(q) $(x \sim (y \downarrow z)) + x$

(r) $x \sim yz \sim z$

$$(s) \quad x \vee y(x \sim z)$$

$$(t) \quad (x \downarrow y)(y \rightarrow z)$$

$$(u) \quad x \mid (yx \sim z)$$

$$(v) \quad (x \downarrow y)(y \downarrow z)$$

$$(w) \quad x \mid ((y \sim z) + z)$$

$$(x) \quad (x \mid y) \sim (y \downarrow z)$$

$$(y) \quad x \rightarrow ((x \vee y) \rightarrow z)$$

$$(z) \quad (x \vee (x \downarrow y)) \mid z$$